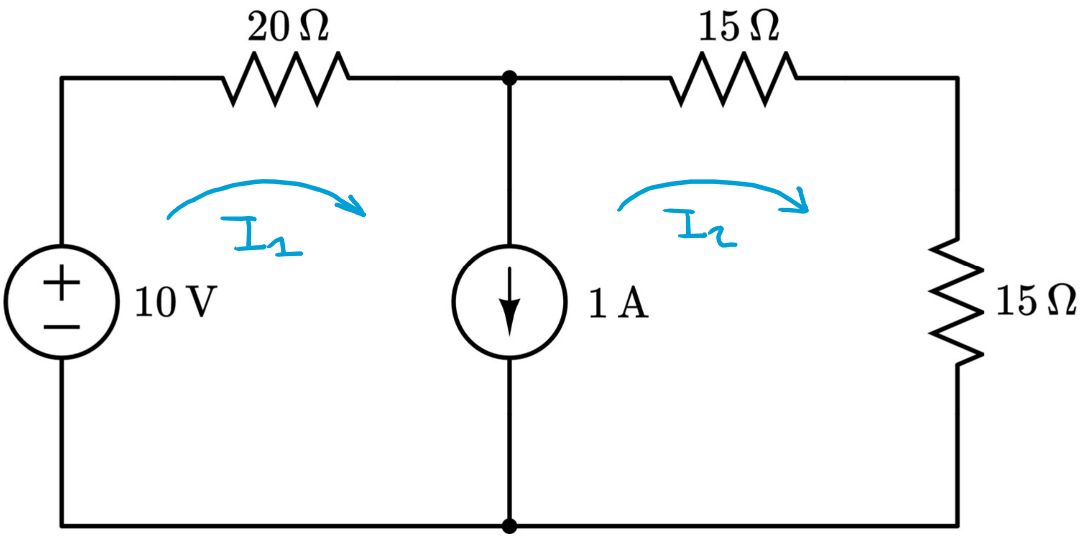
Practice Quiz Five Solution: Mesh Current Analysis (Easy)

For the circuit shown below, use mesh current analysis to define and solve for each mesh current.

Step One: Define each mesh current, identify supermesh



Supermesh: $I_1 - I_2 = 1$

Step Two: KVL following the Supermesh:

$$20I_1 + 15I_2 + 15I_2 = 10$$

$$20I_1 + 30I_2 = 10$$

Step Three: Formulate 2x2 Matrix

$$\begin{bmatrix} 1 & -1 \\ 20 & 30 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix} \qquad \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 20 & 30 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ -\frac{1}{5} \end{bmatrix} A = \begin{bmatrix} 0.8 \\ -0.2 \end{bmatrix} A$$

Self-Grading Rubric	
Clearly name and label each mesh	+2 points
Correct KVL equation following the supermesh	+2 points
Correct supermesh constraint equation	+3 points
Correct answer for i_1 and i_2	+3 points